# SNA R2: Fundamental Network Concepts

**No R in this script, just concepts**

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2025-11-14

This Quarto file contains an introduction to fundamental network and SNA concepts. I expand on all these concepts and discuss them in more detail in later scripts. All SNA4M R Quarto scirpts can be found at https://github.com/jibe4fun/sna4m/tree/R-Code. You can also query all the lectures, R scripts and other material for this course on the [SNA4M ChatBot] (https://poe.com/SNA4M). If you are interested in predictive analytics and machine learning R code, please refer to my PAML4M book R companion in my GitHub site https://github.com/jibe4fun/paml4m/tree/R-Code. If you are interested in Predictive Analytics and Machine Learning for Managers, please refer to my PAML4M book Predictive Analytics and Machine Learning for Managers or chat with my chatbot associated with this book at PAML4M ChatBot.

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**Notes:**

1. I include some R code in this script to render outputs or graphs to complement the conceptual explanations provided in this script. All scripts with the comment '# Illustration' are meant to display graphs and outputs to complement the text narratives, and there is no need to pay attention to the code, unless you want to.
2. Also, sometimes I use the parameter '#| eval: false' inside some R code sections marked with '{r}', which simply prevents the script in that chunk from executing when you render the script into a document, which I use to avoid cluttering the output with unnecessary displays, help information, notifications, etc. To explore these commands, simply go to the respective script line and run it with 'Ctrl-Enter'.

## Network Representations

A **network** is a collection of **actors** and their respective **ties**. The graph terms for actors and ties are **vertices** and **edges**. Vertices are sometimes referred to as **nodes**. Any two vertices form a **dyad**, which is the basic unit of analysis in SNA. One excellent property of SNA and graph analytics methods is that the data can be represented and analyzed visually or quantitatively.

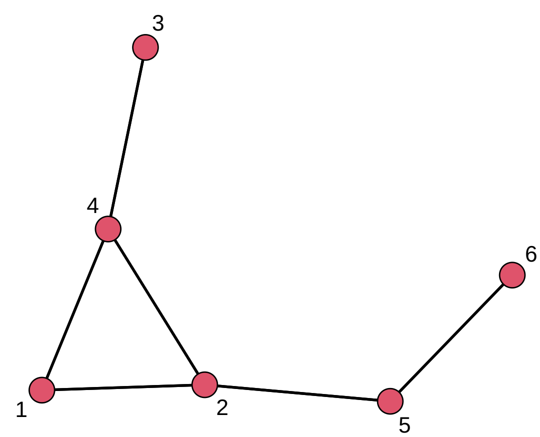
A **visual** representation of graphs is called a **sociogram** in SNA and a **graph** in graph theory.

```
# Illustration

library(statnet) # See package installation instructions below

soc.mat <- rbind(c(0, 1, 0, 1, 0, 0),
                 c(1, 0, 0, 1, 1, 0),
                 c(0, 0, 0, 1, 0, 0),
                 c(1, 1, 1, 0, 0, 0),
                 c(0, 1, 0, 0, 0, 1),
                 c(0, 0, 0, 0, 1, 0))

gplot(soc.mat, displaylabels = T, gmode = "graph")
```




A graph can also be represented as a **quantitative** matrix called a **sociomatrix** (also called an **adjacency matrix**), which makes the power of matrix algebra available to graph analytics and SNA. Cell values contain a **1** if the two related vertices have a tie (for **binary** networks), a **value** representing the weight or strength of the tie (for **weighted** networks), or **0** if the vertices don't have a tie.

```
soc.mat
```

```
     [,1] [,2] [,3] [,4] [,5] [,6]
[1,]    0    1    0    1    0    0
[2,]    1    0    0    1    1    0
[3,]    0    0    0    1    0    0
[4,]    1    1    1    0    0    0
[5,]    0    1    0    0    0    1
[6,]    0    0    0    0    1    0
```

**Diagonal values** are often omitted when vertices don't have relationships of interest with themselves (i.e., recursive relationships).

```r
soc.mat.nodiag <- soc.mat # To avoid altering the original matrix
diag(soc.mat.nodiag) <- "" # Replace all diagonals with blanks
print(soc.mat.nodiag, quote = F)
```

```
     [,1] [,2] [,3] [,4] [,5] [,6]
[1,]      1    0    1    0    0
[2,] 1         0    1    1    0
[3,] 0    0         1    0    0
[4,] 1    1    1         0    0
[5,] 0    1    0    0         1
[6,] 0    0    0    0    1
```

## Vertex Attributes

Vertices can have **attributes.** Some **vertex** attributes are **collected** and some are **measured**:

- **Quantitative:** e.g., age, income, education, etc.
- **Categorical:** e.g., gender, profession, nationality, etc.

Some **vertex** attributes are **computed** with network metrics:

- **Quantitative:** e.g., degree centrality, betweenness centrality, etc.
- **Categorical** or **Membership:** e.g., isolation, clique membership, cluster, community, etc.

It is this ability to compute vertex metrics, in addition to the measured attributes that provides the **enhanced analytical power** of SNA methods.

## Edge Attributes

Edges can also have **attributes.** Some **edge** attributes are **collected** and some are **measured**:

- **Quantitative:** e.g., time working together with a colleague, email exchange volume with a friend, number of courses taken together with a peer, etc.
- **Binary:** e.g., friends with a colleague ( yes/no), LinkedIn contact with someone (yes/no), X follower (yes/no) or follwed by (yes/no), etc.

Some edge attributes are **computed** from network metrics:

- **Quantitative:** e.g., distance to another member, paths to a colleague, walks between two members, etc.
- **Binary:** e.g., is a tie to a member reciprocal, are two colleagues co-members of a clique, group or community, etc.

It is this ability to collect, measure, and compute edge metrics that provides the **enhanced analytical power** of SNA methods.

## Other Attributes

In addition to vertex and edge attributes, networks possess many other attributes at various levels, such as:

- **Sub-Group Level:** Ego/alter network membership, clique co-membership, cluster grouping, community membership, etc.
- **Network Level:** Structure, hierarchy, centralization, isolation, size, number of components, etc.

## Types of Networks

There are multiple types of networks based on several characteristics, such as:

- **Topology:** that is, the configuration of the network, such as star, ring, hierarchical, fully connected, etc.
- **Binary** (Yes/No, 0/1) vs. **Weighted** (i.e., edges have values):
  - **Binary** networks are best for visualization and for computing node metrics, such as centralities. A binary network contains ties between members that have a relationship (e.g., are friends) and no tie if the actors don't have a relationship (e.g., not friends).
  - **Weighted** or **valued** networks are best for quantitative analysis (e.g., clustering).
  - **Dichotomized** networks are those extracted from a weighted network. We often need to convert a weighted network into a binary network. We do that by creating a binary network containing a tie when the weight exceeds a certain **threshold** (i.e., **cut**) of interest.
- **Undirected** vs. **Directed**:
  - An **undirected** network is one in which the relationships does not have a direction. For example, how long have 2 people known each other has no direction because the value is the same for either person, consequently, the ties don't have arrowheads. Undirected networks yield **symmetrical** sociomatrices in which the values above the diagonal are the same as those below the diagonal.
  - **A directed** network is one in which the direction of the relationship matters. For example, how much person A likes person B is not necessarily the same as how much person B likes person A. In this case, how much A likes B is represented as a tie with an arrow going from A to B. Directed networks yield **asymmetrical** sociomatrices in which the values above the diagonal are not necessarily the same as those below the diagonal. The relationship in an asymmetric sociomatrix is specified from row to column (e.g., how much the row member likes the column member).
- **Simplex** vs. **Multiplex**: A simplex network is one representing a single relationship (e.g., friendship). A multiplex network contains two or more layers, one for each relationship represented (e.g., friendship, advice, supervision, etc.).
- **1-Mode** vs. **2-Mode**: In a **1-mode** network, the row actors are the same as the column actors (e.g., a group of classmates), and the cells represent the relationship between row and column actors. In a 2-mode network the row and column actors are different (e.g., rows are students and the columns are courses taken by students). A **2-mode** network can be be used to create (i.e., project) 2 different **1-mode** networks, one representing row x row members (e.g., student x student), with the cells representing some commonality calculated for the row pairs (using sums, averages, differences or correlation), and one representing the column x column entities (e.g., course x course).

## Useful Network Topologies

A network topology is the physical or logical arrangement of vertices and edges within the network. These are useful when comparing networks or when evaluating properties of networks. For example, a star topology has one actor in the center with all other actors connected to it, but not connected to each other. This topology is important because it represents the most centralized network for a given network size. These topologies typically used for computations of network attributes, e.g., network centralization. These are some popular topologies:

- **Fully Connected:** every vertex is connected to every other vertex. The distance from a member to any other member is exactly one tie. A fully connected network has the maximum density possible for a network of a given size.
- **Ring:** a vertex is connected to the next vertex, which in turn is connected to the next vertex, and so on, until the last vertex is connected to the first vertex, closing the loop.
- **Star:** it contains one central vertex with all other vertex connected to it, but not connected to any other vertex. A star network of n vertices has one central vertex with a degree centrality of n-1 and the remaining vertices with a degree centrality of 1.
- **Disconnected:** when there are no paths to connect some members to other members, the network is said to be disconnected. A connected network has one component, which is the actual network. The members have paths to all other members, although the network may not be fully connected. A disconnected network has multiple components. Components are disconnected from other components.
- **Clustered:** all members are connected in triads. A member can be connected with many triads, but it must be connected to at least one triad. It is essentially a network with nothing but triangles.
- **Hierarchical:** it is like a collection of more than one star networks and they all have directed ties. Every member has either no or many child vertices. The vertices without child vertices are the terminal vertices. Every vertex has only one parent vertex, except for the top-most vertex, which has no parent. The best way to visualize a hierarchical network is to think of it as an organizational chart, that is every one has subordinates, except the employees at the lowest level, and every one has a boss, except the top boss.

## Network Analysis Methods

There are two main types of network methods: quantitative and visual. In addition, there are a number of transformations that can be applied to network data to unleash the great power of the various methods.

- **Quantitative:** this method relies on matrix manipulations and quantitative and statistical methods. Relationships in a network can be represented as binary or weighted inside a sociomatrix, which can then be inputed into the various quantitative methods available.
- **Visual:** sociomatrices can be graphed and then analyzed using various visual methods. For example: vertices can be sized based on their centrality or other attribute; edges can be heavier or lighter depending on weighted edge attributes; communities and clusters can be colored and reported in informational visual graphs.
- **Transformations:** these are not methods per se, but a clever transformation can lead into unique insights from the data. We will discuss many of these in later chapters. Some examples of useful transformations include:
  - **Matricizing:** it involves converting non-network or non-matrix data into matrix data. A simple example is when you have a number of actors of different ages and having different incomes. You could use this data to construct a matrix that shows age or income differences between members. You can not only use differences, but you can also use other values like sums, averages, etc. Another example is to convert text to relations. For example, in the Network of Thrones example, analysts computed how many time every 2 actors in the novel appeared together in the text no more than 15 words appart, and used this count as a proximity value between them.

- **2-Mode to 1-Mode Projections:** for example, if you have a 2-mode network containing students (rows) by courses (columns), with a 1 if the row student took the column course. You can then project this into a 1-mode student x student network, with cells containing how many times the row and column students took the same course.
- **Aggregate, dichotomize, group, etc.:** sky is the limit when it comes to transformations.

## Levels of Network Analysis

One of the strengths of network analysis is that there are many levels of analysis offering a rich variety of perspectives, such as:

- **Vertex (Node) Level:** collect or compute and analyze various node level and ego network attributes (centrality, isolates, etc.), such as:
  - **Actor Prominence:** degree, betweenness or some other measure of centrality
  - **Vertex Bridge:** when the only way to establish a path between two actors is through another actor, this actor is called a vertex bridge
  - **Isolates:** vertices with no connections to others
  - **Ego Network:** one actor's ties to neighboring actors (1-degree, i.e., 1 tie away, or n-degree, i.e., up to n ties away)
  - **Structural Hole:** an actor with many ties to actors that are not tied to each other (i.e., stars)
- **Edge (Tie) Level:** collect or compute and analyze relationship data (strong/weak ties, tie bridges, reciprocity, transitivity, etc.), such as:
  - **Edge Bridge:** when the only way to establish a path between two vertices is through an edge between them, this edge is called an edge bridge – the two nodes in an edge bridge are also vertex bridges
  - **Path:** a sequence of adjacent vertices and edges connecting two nodes. Sometimes there is a single path between two nodes; other times there are multiple paths.
  - **Distance:** the number of ties in the shortest path between 2 vertices, also known as the Geodesic. Note that if the network is disconnected into more than one component, the distance between some members will be indeterminate (i.e., the distance between 2 members across unconnected components is infinite).
  - **Reciprocity:** it applies to directed networks only. : if A –> B and B –> A, the tie is reciprocal. More generally, the tie between two actors can be either: none existent, directional or reciprocal.
- **Sub-Group Level:** identify and analyze sub-groups, such as:
  - **Cliques:** these are N vertices fully connected (3-clique, 4-clique, etc.)
  - **Sub-Graphs:** a focal part of a graph (e.g., a subgroup of members of the same gender or country)
  - **Components:** fully connected sub-graphs, disconnected from other sub-graphs
  - **Clusters:** groups of nodes with close proximity or similarity
  - **Communities:** similar to clusters. More specifically, they are groups of actors with many ties within the community and sparse ties to members in other communities.
  - **Triads (Transitivity):** 3-actor cliques A –> B, A –> C and B –> C (i.e., forming a triad or triangle). Related to this, if A –> B and B –> C and A –> C, means that the relationship between A and C is transitive and A, B and C form a triad cluster. Note: A network with high transitivity (many triads) has a high clustering coefficient and, consequently, it is low on structural holes (few stars)
- **Network Level:** compute and analyze key network structural parameters, such as:

- **Network Size:** Number of vertices. Some social networks are small and specific (e.g., 12-member team) and can be analyzed as a single network. Sometimes you have many teams, and you want to predict team outcomes (e.g., 50 small teams of average size of 5). Sometimes you have very large groups (e.g., communities of practice, social media groups) with thousands of nodes; visualization and data collection are more difficult.
  - **Network Density:** Number of ties (edges), or the proportion of ties relative to the maximum number of possible ties for a network of a given size. Some networks are sparse (few ties/lots of weak ties) and some are dense (many ties/lots of strong ties).
  - **Diameter:** is largest distance (geodesic) between actors in the network
  - **Other:** Centralization, Isolation, Clustering, Hierarchy, Structure, etc.

- **Multiplex Networks:** extract simplex network layers from a multiplex network for each relationship of interest and analyze their relationships or associations separately (e.g., correlation, regression):
  - **Vertex Predictors** (gender, centrality) –> **Vertex Outcomes** (success)
  - **Edge Relationships** (supervisor, advice) –> **Edge Outcomes** (friendship)
  - **Mixed** (Vertex –> Edge; Edge –> Vertex)